

The effort was directed towards second-generation cellular systems.

In 1990, a second frequency band was specified. This band included two domains—1710-1785 MHz and 1805-1880 MHz, i.e., twice 75 MHz; three times as much as the primary 900MHz band.

Digital enhanced cordless telecommunication (DECT). DECT is a type of PCS system. DECT standard was developed by European Telecommunication Standards Institute (ETSI) for wireless PABX data LAN applications that represent closed environments requiring minimal open cordless access, since it was essential that products from different vendors not only coexist but interwork with each other.

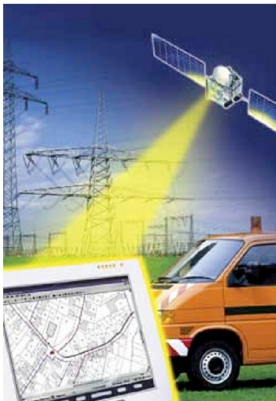
DECT system has a TDMA/TDD frame structure with 24 slots that are equally allocated for downlink and uplink operation. DECT specifies both simplex (half-slots) and duplex (full slot) operation. Higher data rates are achieved by utilising multilevel modulation. The basic modulation scheme is a two-level Gaussian filled frequency-shift keying (GFSK), which is supplemented with 8-level modulation scheme leading to as high as 2.88 Mbps per carrier.

GSM

Global system for mobile communications (GSM) was developed by the Groupe Special Mobile, which was an initiative of the Conference of European Post and Telecommunications (CEPT) administrations. GSM was first devised as a cellular system in a specific 900MHz band called the primary band. This primary band includes two sub-bands of 25 MHz each, 890-915 MHz and 935-960 MHz.

GSM systems like Iridium, Globalstar and ICO use constellations of low-earth orbit (LEO) or medium-earth orbit (MEO) satellites and operate as overlay networks for existing cellular and PCS networks. Using dual-mode, these extend the coverage to any and all locations on the earth's surface.

International Mobile Telecommunication-2000 (IMT-2000) is a standard developed by ITU for 3G.



GPS—a reliable navigational aid anywhere on the earth

It ensures global mobility in terms of global seamless roaming and service delivery. An appreciation of the role of numbering and identities in mobility management, international roaming, call delivery, and billing and charging is important in understanding the operation of mobile and personal communication networks.

Personal communication satellite service (PCSS) uses LEO satellite repeaters incorporating QPSK modulation and both FDMA and TDMA.

The main advantages of GSM are international roaming (in harmony with ISDN principles assuring interworking between ISDN and GSM) and features like privacy and encryption, frequency hopping, discontinuous transmission and short message service. Other facilities include call forwarding, barring, waiting, hold and teleconferencing.

The basic architecture comprises a network sub-system, base station sub-system, mobile stations, and system interworking and interfaces.

A subscriber identity module (SIM) is required to activate and operate a GSM terminal. The SIM may be contained within the mobile station or

it may be a removable unit that can be inserted by the user in his mobile set.

New developments along the way

Before we proceed to evolution from 1G to 4G, let me touch upon the new developments that took place in 1G to 4G.

Global positioning system (GPS). GPS is a reliable navigational aid available anywhere on the earth, operating in all weather conditions 24 hours a day. It can be used by marine, airborne and land users. GPS technology was developed in 1983.

GPS consists of three segments:

Space segment. GPS consists of 24 NAVSTAR satellites along with three spare satellites orbiting at 20,200 km above the earth's

surface in six circular orbital planes with a 12-hour orbital period each. These satellites operate at L1 band (1.575 GHz) continuously broadcasting navigational signals called coarse acquisition code. These codes can be received by anyone for decoding and finding navigational parameters like longitude, latitude, velocity and time.

Control segment. It consists of a master control station (MCS) and a number of smaller earth stations called monitoring stations located at different places in the world. Monitoring stations track satellites and pass on the measured data to the MCS. The MCS computes satellite parameters (called ephemeris) and sends them back to the satellite, which, in turn, broadcasts to all GPS receivers.

User segment. The user segment consists of all moving and stationary objects with GPS receivers. A GPS receiver is a multi-channel satellite receiver that computes every second its own location and velocity.

Bluetooth. Compared to WLAN technologies, Bluetooth technology aims at so-called *ad-hoc* piconets, which are local-area networks with a very limited coverage and without the